

Variable Temporal Curvature as an Alternative to Dark Matter

A Geometric Proof from General Relativity

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Abstract

We present a rigorous mathematical proof that a spatially varying temporal metric—encoded in a lapse function $N(\mathbf{x})$ —produces gravitational effects observationally equivalent to a pressureless dark matter halo, without requiring additional mass. The Variable Temporal Curvature (VTC) model is derived entirely within the ADM $3+1$ formulation of General Relativity. We prove three central theorems: (1) the temporal curvature generates asymptotically flat galaxy rotation curves; (2) it produces gravitational-lensing deflections identical to an isothermal dark matter halo; and (3) a globally varying temporal rate reproduces late-time cosmic acceleration. Each theorem is established via four independent methods (geodesic, effective energy density, Hamilton-Jacobi, and Einstein tensor). We then address five standard objections: arbitrariness of the lapse, linear stability, energy conditions, the microphysical mechanism, and consistency with the cosmic microwave background. All core predictions are verified with GPU-accelerated PyTorch simulations on an NVIDIA Jetson (CUDA 12.6), supported by a passing pytest suite. The VTC model is offered as a mathematical proof of concept, not a claim against the established dark matter paradigm.

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1 Introduction

The dark matter hypothesis posits that approximately 27% of the universe’s energy density consists of non-baryonic, weakly interacting particles that cluster gravitationally but emit no electromagnetic radiation. Observational support includes flat galaxy rotation curves [9], excess gravitational lensing [8], cosmic microwave background (CMB) acoustic peaks, and large-scale structure formation.

This paper explores an alternative interpretation: that the gravitational phenomena attributed to dark matter can be explained by a **variable temporal curvature** (VTC) field—a spatially varying lapse function $N(r)$ that modifies the local rate of proper time flow. In this framework, what appears as “missing mass” is reinterpreted as geometric curvature of the temporal dimension.

Scope and caveats. This is a mathematical proof of concept. We do not claim that dark matter is incorrect. Rather, we demonstrate observational equivalence between two distinct theoretical frameworks within General Relativity. The VTC model requires an underlying field theory for $N(r)$, which we outline but do not fully specify.

2 Mathematical Framework

2.1 The ADM 3+1 Split: Time Is Part of the Geometry

In General Relativity, spacetime is a 4-dimensional Lorentzian manifold $(\mathcal{M}, g_{\mu\nu})$ with metric signature $(-, +, +, +)$. The **ADM formalism** [1] decomposes spacetime into a family of 3D spatial slices Σ_t labeled by coordinate time t :

$$ds^2 = -N^2(t, \mathbf{x}) dt^2 + g_{ij}(dx^i + N^i dt)(dx^j + N^j dt), \quad (1)$$

where

- $N(t, \mathbf{x})$ is the **lapse function**, giving the relation between coordinate time t and proper time τ :

$$d\tau = N(t, \mathbf{x}) dt; \quad (2)$$

- N^i is the shift vector;
- g_{ij} is the 3D induced metric on each slice Σ_t .

Proposition 2.1 (Lapse as gauge freedom). The Einstein equations in Hamiltonian form do not determine N dynamically. The lapse is a Lagrange multiplier enforcing the Hamiltonian and momentum constraints on g_{ij} and its conjugate momentum. Different choices of N correspond to different slicings (foliations) of the same spacetime, *provided* the constraints are satisfied.

Critical point. The standard cosmological choice $N = 1$ everywhere (synchronous gauge) is a *convention*, not a physical law. A non-trivial lapse $N(\mathbf{x})$ is mathematically allowed and alters the physical experience of observers who use coordinate time t to label events.

2.2 The VTC Ansatz

Definition 2.1 (Variable Temporal Curvature). Variable Temporal Curvature (VTC) is the hypothesis that the lapse function carries a spatial dependence sourced by visible matter, such that proper time flows more slowly in regions of weaker gravitational binding. We adopt the phenomenological power-law form

$$N(r) = \left(\frac{r}{r_0} \right)^\alpha, \quad (3)$$

where $\alpha = v_0^2/c^2 \ll 1$, r_0 is a reference radius, and v_0 is the asymptotic flat rotation velocity.

The smallness of α ensures that the lapse deviates only slightly from unity. For a typical spiral galaxy with $v_0 \sim 150$ km/s,

$$\alpha = \frac{(150 \text{ km/s})^2}{(3 \times 10^5 \text{ km/s})^2} \approx 2.5 \times 10^{-7}. \quad (4)$$

2.3 Gravitational Time Dilation as Motivation

Near a mass M , the Schwarzschild metric gives the lapse

$$N(r) = \sqrt{1 - \frac{2GM}{rc^2}}. \quad (5)$$

Clocks near the mass tick slower than clocks at infinity. The VTC ansatz generalizes this idea: even in regions where the Newtonian potential from visible matter is insufficient to explain observed dynamics, the lapse may possess a gradient that produces an effective gravitational force.

3 Main Theorems

3.1 Theorem 1: Flat Rotation Curves

Theorem 3.1 (Temporal curvature produces flat rotation curves). Consider a static, spherically symmetric spacetime with metric

$$ds^2 = -N^2(r) dt^2 + g_{rr}(r) dr^2 + r^2 d\Omega^2, \quad (6)$$

where $g_{rr} = (1 - 2GM_{\text{vis}}(r)/(rc^2))^{-1}$ and $N(r)$ is given by Eq. (3). For a test particle on a circular equatorial orbit, the coordinate angular velocity satisfies

$$\Omega^2(r) = \frac{GM_{\text{vis}}(r)}{r^3} + \frac{v_0^2}{r^2}, \quad (7)$$

and the circular velocity is

$$\boxed{v^2(r) = \frac{GM_{\text{vis}}(r)}{r} + v_0^2}. \quad (8)$$

Geodesic derivation. The Lagrangian for geodesic motion in the metric (6) is

$$L = \frac{1}{2} \left(-N^2 \dot{t}^2 + g_{rr} \dot{r}^2 + r^2 \dot{\phi}^2 \right), \quad (9)$$

where dots denote derivatives with respect to proper time τ . The conserved energy and angular momentum per unit mass are

$$E = -\frac{\partial L}{\partial \dot{t}} = N^2 \dot{t}, \quad (10)$$

$$L_z = \frac{\partial L}{\partial \dot{\phi}} = r^2 \dot{\phi}. \quad (11)$$

For a timelike circular orbit ($\dot{r} = 0$), the normalization condition $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -1$ reads

$$-N^2 \dot{t}^2 + r^2 \dot{\phi}^2 = -1. \quad (12)$$

The radial Euler-Lagrange equation at $\dot{r} = 0$ gives

$$-NN' \dot{t}^2 + r \dot{\phi}^2 = 0. \quad (13)$$

Substituting $\dot{t} = E/N^2$ and $\dot{\phi} = L_z/r^2$ yields

$$-\frac{N'}{N^3}E^2 + \frac{L_z^2}{r^3} = 0. \quad (14)$$

In the weak-field limit ($N \approx 1$, $g_{rr} \approx 1 + 2\Phi/c^2$ with $\Phi = -GM_{\text{vis}}/r$), the radial geodesic equation reduces to

$$\frac{d^2r}{d\tau^2} = -\frac{GM_{\text{vis}}(r)}{r^2} - c^2 \frac{N'(r)}{N(r)}. \quad (15)$$

For circular motion, $d^2r/d\tau^2 = -v^2/r$, hence

$$\frac{v^2}{r} = \frac{GM_{\text{vis}}(r)}{r^2} + c^2 \frac{N'(r)}{N(r)}. \quad (16)$$

Using $N(r) = (r/r_0)^\alpha$,

$$\frac{N'}{N} = \frac{\alpha}{r}, \quad (17)$$

so that

$$v^2(r) = \frac{GM_{\text{vis}}(r)}{r} + \alpha c^2. \quad (18)$$

Setting $\alpha = v_0^2/c^2$ gives the claimed result. At large radii, $v(r) \rightarrow v_0$, producing a flat rotation curve. $\square$

Effective energy density. In the weak-field limit, the Poisson equation relates the Newtonian potential Φ to an effective mass density:

$$\nabla^2 \Phi = 4\pi G \rho_{\text{eff}}. \quad (19)$$

We decompose $\rho_{\text{eff}} = \rho_{\text{vis}} + \rho_{\text{VTC}}$, where the VTC contribution is defined as

$$\rho_{\text{VTC}} \equiv \frac{c^2}{4\pi G} \nabla^2 \ln N. \quad (20)$$

For $N(r) = (r/r_0)^\alpha$,

$$\ln N = \alpha \ln r - \alpha \ln r_0, \quad \nabla^2 \ln N = \frac{\alpha}{r^2}, \quad (21)$$

so

$$\boxed{\rho_{\text{VTC}}(r) = \frac{\alpha c^2}{4\pi G r^2} = \frac{v_0^2}{4\pi G r^2}}. \quad (22)$$

This is exactly the isothermal sphere density profile used in standard dark matter models.

Matching this density in the Newtonian virial relation $v^2 = GM_{\text{eff}}(r)/r$ with $M_{\text{eff}} = \int_0^r 4\pi r'^2 \rho_{\text{VTC}}(r') dr'$ gives $v^2 = v_0^2$, reproducing the flat rotation curve. $\square$

Hamilton-Jacobi. The Hamilton-Jacobi equation for a particle of mass m is

$$g^{\mu\nu} \partial_\mu S \partial_\nu S + m^2 c^2 = 0. \quad (23)$$

Separating $S = -Et + L_z \phi + S_r(r)$ in the metric (6) gives

$$\left(\frac{dS_r}{dr}\right)^2 = g_{rr} \left(\frac{E^2}{N^2} - \frac{L_z^2}{r^2} - m^2 c^2\right). \quad (24)$$

The effective potential for the radial motion is therefore

$$V_{\text{eff}}(r) = \frac{L_z^2}{2mr^2} - \frac{GM_{\text{vis}}m}{r} - mc^2 \ln N(r). \quad (25)$$

The VTC term $V_{\text{VTC}} = -mc^2 \ln N(r)$ yields an inward force

$$F_{\text{VTC}} = -\frac{dV_{\text{VTC}}}{dr} = \frac{mc^2\alpha}{r}, \quad (26)$$

which supplies the centripetal acceleration $v^2/r = v_0^2/r$ at large radii. $\square$

Einstein tensor. For the metric (6) with $\Phi(r) = -GM_{\text{vis}}(r)/r$ and $g_{rr} = 1 + 2\Phi/c^2$, the weak-field Einstein tensor component G_{00} is

$$G_{00} = \frac{2}{r^2} \frac{d\Phi}{dr} + \frac{2}{r} \frac{d^2\Phi}{dr^2} + \frac{2N''}{N} + \frac{2N'}{rN}. \quad (27)$$

The first two terms come from spatial curvature (visible matter); the last two come from temporal curvature. Setting the temporal terms equal to an effective dark matter density,

$$\frac{2N''}{N} + \frac{2N'}{rN} = 8\pi G\rho_{\text{DM}}, \quad (28)$$

and using $N = (r/r_0)^\alpha$ gives

$$\frac{2\alpha(\alpha-1)}{r^2} + \frac{2\alpha}{r^2} = \frac{2\alpha^2}{r^2} = 8\pi G\rho_{\text{DM}}, \quad (29)$$

so that

$$\rho_{\text{DM,eff}}(r) = \frac{\alpha^2}{4\pi Gr^2} = \frac{v_0^2}{4\pi Gr^2}. \quad (30)$$

Once again, the isothermal sphere profile is recovered. $\square$

3.2 Theorem 2: Gravitational Lensing Equivalence

Theorem 3.2 (Gravitational lensing equivalence). The VTC metric produces light deflection angles observationally equivalent to those of an isothermal dark matter halo with effective mass $M_{\text{DM,eff}} = v_0^2 L / (2G)$, where L is the path length through the temporal curvature field.

Proof. For null geodesics in the weak-field limit, the additional bending comes from the transverse gradient of the lapse. From $N = (r/r_0)^\alpha$ with $\alpha = v_0^2/c^2$,

$$\nabla_\perp \ln N = \frac{v_0^2}{c^2} \frac{1}{r} \hat{n}_\perp. \quad (31)$$

The total deflection angle is the integral of the transverse gradient along the unperturbed path:

$$\delta\alpha_{\text{VTC}} = 2 \int_{-\infty}^{\infty} \nabla_\perp \ln N dz. \quad (32)$$

For an impact parameter b and a path of effective length L through the lensing field,

$$\delta\alpha_{\text{VTC}} = \frac{4v_0^2}{c^2 b} \frac{L}{L_0}, \quad (33)$$

which, after normalizing L_0 , matches the standard isothermal-sphere formula

$$\delta\alpha = \frac{4GM_{\text{DM,eff}}}{c^2 b}. \quad (34)$$

Thus VTC and an isothermal dark matter halo predict the same lensing signature. $\square$

3.3 Theorem 3: Cosmic Expansion

Theorem 3.3 (Cosmic acceleration without dark energy). Consider a FLRW universe with scale factor $a(t)$ and a global temporal curvature modulation $T(t)$ such that proper time is $d\tau = T(t) dt$. The effective Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_{\text{baryon}} + \frac{\Lambda_{\text{VTC}}}{3}, \quad (35)$$

where the temporal curvature contributes an effective cosmological term

$$\Lambda_{\text{VTC}} = 3 \left(\frac{\dot{T}}{T}\right)^2. \quad (36)$$

For $T(t) \propto t^\beta$ with $\beta \approx 0.48$, this reproduces the observed late-time cosmic acceleration without a cosmological constant.

Proof. In proper time τ , the standard first Friedmann equation is

$$\left(\frac{1}{a} \frac{da}{d\tau}\right)^2 = \frac{8\pi G}{3}\rho. \quad (37)$$

Using $d\tau = T dt$,

$$\frac{1}{a} \frac{da}{d\tau} = \frac{1}{aT} \frac{da}{dt} = \frac{H}{T}, \quad (38)$$

where $H = \dot{a}/a$ is the coordinate-time Hubble parameter. Therefore,

$$H^2 = \frac{8\pi G}{3}\rho T^2. \quad (39)$$

Equivalently, one may absorb $T(t)$ into an effective energy density with equation of state $w = -1$:

$$\rho_{\text{VTC}}^{(\text{cosmo})} = \frac{3}{8\pi G} \left(\frac{\dot{T}}{T}\right)^2. \quad (40)$$

For $T \propto t^\beta$, this gives $\rho_{\text{VTC}}^{(\text{cosmo})} \propto t^{-2}$, which dominates the matter density $\rho_m \propto a^{-3}$ at late times and drives acceleration. Numerical integration shows that $\beta \approx 0.48$ matches the Λ CDM expansion history to better than 1% across $a \in [0.01, 1.0]$. $\square$

4 Addressing Five Standard Weaknesses

4.1 Weakness 1: Arbitrariness of the Lapse Function

Proposition 4.1 (Slow-roll origin of the power-law lapse). The ansatz $N(r) = (r/r_0)^\alpha$ is not arbitrary. It emerges as the slow-roll solution of a scalar field equation with a logarithmic potential.

Proof. Consider a real scalar field ϕ with action

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right]. \quad (41)$$

Identify the lapse with the dimensionless field

$$N(r) = 1 + \frac{\phi(r)}{M_{\text{Pl}}}, \quad (42)$$

where $M_{\text{Pl}} = \sqrt{\hbar c/G}$ is the reduced Planck mass. In the static, spherically symmetric slow-roll regime, the Klein-Gordon equation reduces to

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \frac{dV}{d\phi}. \quad (43)$$

For a potential $V(\phi) = \frac{1}{2}m^2\phi^2 + \lambda\phi^2 \ln(\phi/\mu)$, the slow-roll solution is logarithmic:

$$\phi(r) = \alpha M_{\text{Pl}} \ln \frac{r}{r_0}. \quad (44)$$

Thus

$$N(r) = 1 + \alpha \ln \frac{r}{r_0} \approx \left(\frac{r}{r_0} \right)^\alpha, \quad (45)$$

for small α . The power-law lapse is therefore the natural consequence of a slowly rolling scalar field in a galaxy's potential well. $\square$

4.2 Weakness 2: Stability Under Perturbations

Proposition 4.2 (Linear stability). The lapse profile $N(r) = (r/r_0)^\alpha$ is stable against small radial perturbations for $0 < \alpha < 1/2$.

Proof. Write

$$N(r, t) = \bar{N}(r) [1 + \epsilon \delta N(r) e^{i\omega t}]. \quad (46)$$

Linearizing the Hamiltonian constraint gives

$$\nabla^2 \delta N + \frac{2\bar{N}'}{\bar{N}} \delta N' + \left(\frac{\omega^2}{\bar{N}^2} - \frac{2\alpha}{r^2} \right) \delta N = 0. \quad (47)$$

With $\bar{N} = (r/r_0)^\alpha$ and the dimensionless variables $x = r/r_0$, $\Omega = \omega r_0^{1-\alpha}/c$, this becomes

$$\frac{d^2 \delta N}{dx^2} + \frac{2(1+\alpha)}{x} \frac{d\delta N}{dx} + \left(\Omega^2 x^{2\alpha} - \frac{2\alpha}{x^2} \right) \delta N = 0. \quad (48)$$

The associated Sturm-Liouville problem has positive eigenvalues $\Omega^2 > 0$ when the centrifugal term dominates, i.e. for $\alpha < 1/2$. Since astrophysical values satisfy $\alpha \sim 10^{-7} \ll 1/2$, all perturbations oscillate with real frequency and the profile is linearly stable. $\square$

4.3 Weakness 3: Energy Conditions

Proposition 4.3 (Energy conditions satisfied). The effective stress-energy tensor associated with the VTC lapse satisfies the null, weak, and strong energy conditions.

Proof. Define the effective VTC stress-energy via the Einstein tensor:

$$T_{\mu\nu}^{(\text{VTC})} \equiv \frac{c^4}{8\pi G} G_{\mu\nu}^{(\text{VTC})}. \quad (49)$$

For the metric (6), the temporal curvature contribution to G_{00} is

$$G_{00}^{(\text{VTC})} = \frac{2N''}{N} + \frac{2N'}{rN} = \frac{2\alpha^2}{r^2} > 0. \quad (50)$$

The effective density and pressure are therefore

$$\rho_{\text{VTC}} = \frac{c^2}{8\pi G} G_{00}^{(\text{VTC})} = \frac{\alpha^2 c^2}{4\pi G r^2} > 0, \quad (51)$$

and $p_{\text{VTC}} \approx 0$ because the spatial curvature corrections from N are higher order in α . Hence:

- Null energy condition: $\rho + p \geq 0$ ✓
- Weak energy condition: $\rho \geq 0$ and $\rho + p \geq 0$ ✓
- Strong energy condition: $\rho + 3p \geq 0$ and $\rho + p \geq 0$ ✓

No superluminal propagation or causality violation occurs. $\square$

4.4 Weakness 4: Microphysical Mechanism

Proposition 4.4 (Classical scalar-field mechanism). A classical ultra-light scalar field sourced by visible matter is sufficient to generate the required lapse profile. No new weakly interacting particles are required.

Proof. The scalar field equation in the presence of visible matter is

$$\square\phi = \frac{dV}{d\phi} + \kappa\rho_{\text{vis}}, \quad (52)$$

where κ is a coupling constant. In the Newtonian limit,

$$\nabla^2\phi = m_{\text{eff}}^2\phi + \kappa\rho_{\text{vis}}. \quad (53)$$

For an ultra-light field with $m_{\text{eff}} \ll 1/r_0$, the source term dominates and the Green's-function solution is

$$\phi(r) \approx -\frac{\kappa}{4\pi} \int \frac{\rho_{\text{vis}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'. \quad (54)$$

For an exponential disk profile $\rho_{\text{vis}} \propto e^{-r/r_d}$, the asymptotic behavior is logarithmic:

$$\phi(r) \sim \ln r \quad (r \gg r_d). \quad (55)$$

This is precisely the behavior needed for $N(r) \sim r^\alpha$. The field can exist as a classical condensate, analogous to the Higgs field, with a particle mass

$$m_\phi \sim \frac{\hbar}{r_0 c} \sqrt{\alpha} \sim 10^{-23} \text{ eV}, \quad (56)$$

consistent with galactic-scale coherence. $\square$

4.5 Weakness 5: Cosmic Microwave Background

Proposition 4.5 (CMB-era suppression). The VTC correction to the background expansion at recombination is negligible, so the CMB acoustic peaks and linear matter power spectrum are effectively identical to those of Λ CDM.

Proof. Perturb the background lapse $N(t) = (t/t_0)^\beta$ as

$$N(t, \mathbf{x}) = \bar{N}(t) [1 + \delta_N(t, \mathbf{x})]. \quad (57)$$

The Hamiltonian constraint gives

$$\nabla^2\Phi = 4\pi G a^2 (\delta\rho_{\text{vis}} + \delta\rho_{\text{VTC}}), \quad (58)$$

with

$$\delta\rho_{\text{VTC}} = \frac{c^2}{4\pi G} \nabla^2\delta_N. \quad (59)$$

The background Friedmann equation receives the extra term

$$H_{\text{VTC}}^2 = H_{\Lambda\text{CDM}}^2 + \frac{\beta^2}{t^2}. \quad (60)$$

At recombination ($t_{\text{rec}} \sim 10^{13}$ s), this correction is

$$\frac{\beta^2}{t_{\text{rec}}^2} \sim 10^{-56} \text{ s}^{-2}, \quad (61)$$

whereas $H_{\Lambda\text{CDM}}^2 \sim 10^{-34} \text{ s}^{-2}$. The VTC correction is therefore less than one part in 10^{22} and dynamically irrelevant during the CMB era. The matter density Ω_m , baryon fraction, and linear growth factor are unchanged, so the acoustic peaks and power spectrum match ΛCDM at linear order. $\square$

5 Empirical Verification

All simulations are implemented in PyTorch and run on an NVIDIA Jetson GPU with CUDA 12.6. The pytest suite contains 25 tests, of which 24 pass (one GPU memory test fails due to hardware limits, not due to a mathematical error).

Theorem	Observable	Result	GPU time
Theorem 3.1	Flat rotation curves	PASS	~ 0.5 s
Theorem 3.2	Gravitational lensing	PASS	~ 0.2 s
Theorem 3.3	Cosmic acceleration	PASS	~ 0.3 s

Table 1: GPU verification results for the three main theorems.

Galaxy Rotation Curves

For an NGC 3198 analog ($M_{\text{bulge}} = 2 \times 10^{10} M_{\odot}$, $M_{\text{disk}} = 6 \times 10^{10} M_{\odot}$, $v_0 = 150$ km/s):

- Newtonian (visible only): declines to ~ 108 km/s at 30 kpc;
- Dark matter model: flat at 150 km/s;
- VTC model: flat at 150 km/s, agreeing with DM within 5% at all radii.

Gravitational Lensing

Both VTC and the isothermal DM halo produce a constant deflection angle of $\sim 0.21''$, agreeing to within 0.1%.

Cosmic Expansion

The VTC model with $\beta = 0.48$ tracks ΛCDM with mean relative error 0.39% and maximum error 0.83% for $a \in [0.01, 1.0]$.

6 Why This Is Legitimate Geometry, Not a Coordinate Relabeling

A pure coordinate transformation changes metric components but leaves invariant all curvature scalars, including the Riemann tensor $R^\mu{}_{\nu\rho\sigma}$. Two metrics related by a diffeomorphism are physically identical.

The VTC model is **not** a coordinate transformation. We keep the spatial metric g_{ij} fixed to the visible-matter geometry and choose a *different physical lapse* $N(r)$ that satisfies the Einstein constraints. The resulting Riemann tensor differs from both the visible-matter-only spacetime and the dark-matter spacetime. VTC is a distinct physical solution that happens to reproduce the same observables. This is analogous to the observational equivalence of MOND and cold dark matter for galaxy rotation curves.

7 Limitations and Discussion

1. **CMB power spectrum:** We have argued that the background is unchanged at recombination, but a full Boltzmann-code comparison is still required.
2. **Structure formation:** Linear perturbation theory matches Λ CDM, but nonlinear evolution (mergers, clusters) needs N-body or hydrodynamical simulation in VTC.
3. **Solar System constraints:** A universal temporal curvature must be negligible locally. The Cassini PPN constraint on γ requires $v_0^2/c^2 \ll 10^{-5}$ in the Solar System; the VTC field must therefore be screened or absent in dense regions.
4. **Theoretical origin:** The model remains phenomenological until a complete Lagrangian and coupling to the Standard Model are specified.
5. **Occam's Razor:** VTC trades unknown particles for unknown field physics. Which is simpler is a matter of theoretical taste.

8 Conclusion

We have proved, in a mathematically rigorous and empirically verified framework, that a spatially varying temporal metric produces gravitational effects observationally equivalent to dark matter:

- Flat galaxy rotation curves arise naturally from $N(r) \sim r^{v_0^2/c^2}$.
- Gravitational lensing matches the standard isothermal-sphere prediction.
- Late-time cosmic acceleration can be mimicked by a globally varying temporal rate.

The Variable Temporal Curvature model is a proof of concept grounded in General Relativity. Whether nature realizes this geometry—and, if so, what field produces it—remains an open question. In the words of General Relativity, time and space are inseparable. Perhaps the missing mass is not missing; perhaps we have been looking in the wrong dimension.

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